AFTERWARD

Clearer discharge documents. Fewer unresolved questions.

Understand what is there. Find what is missing.
Keep the remaining conversation connected.

Google Gemma on AMD | Track 2: Health and Wellbeing

Live demo: afterward-demo.up.railway.app

THE PROBLEM

A correct letter can still leave a practical gap

The patient needs an answer. The ward needs the context.

"Follow the attached rehabilitation programme."

But the attachment is not in the packet.
Or the instruction is present, but unfamiliar.

8.05 million

discharges recorded in Italy in 2024 Episodes across care regimes and outcomes.

14.5 minutes

median summary completion in one hospital study of 7,185 records.

Sources: ISS / SDO 2024; Kleinig et al., 2025. Context only: neither figure measures AfterWard impact.

THE WORKFLOW

Start where the documents already are

One case connects source documents, AI suggestions and the remaining questions.

01 Before discharge

The ward checks the packet, adds missing documents and shares the reviewed bundle.

02 At home

The patient reads simpler wording and looks for answers in their documents.

03 Ask the ward

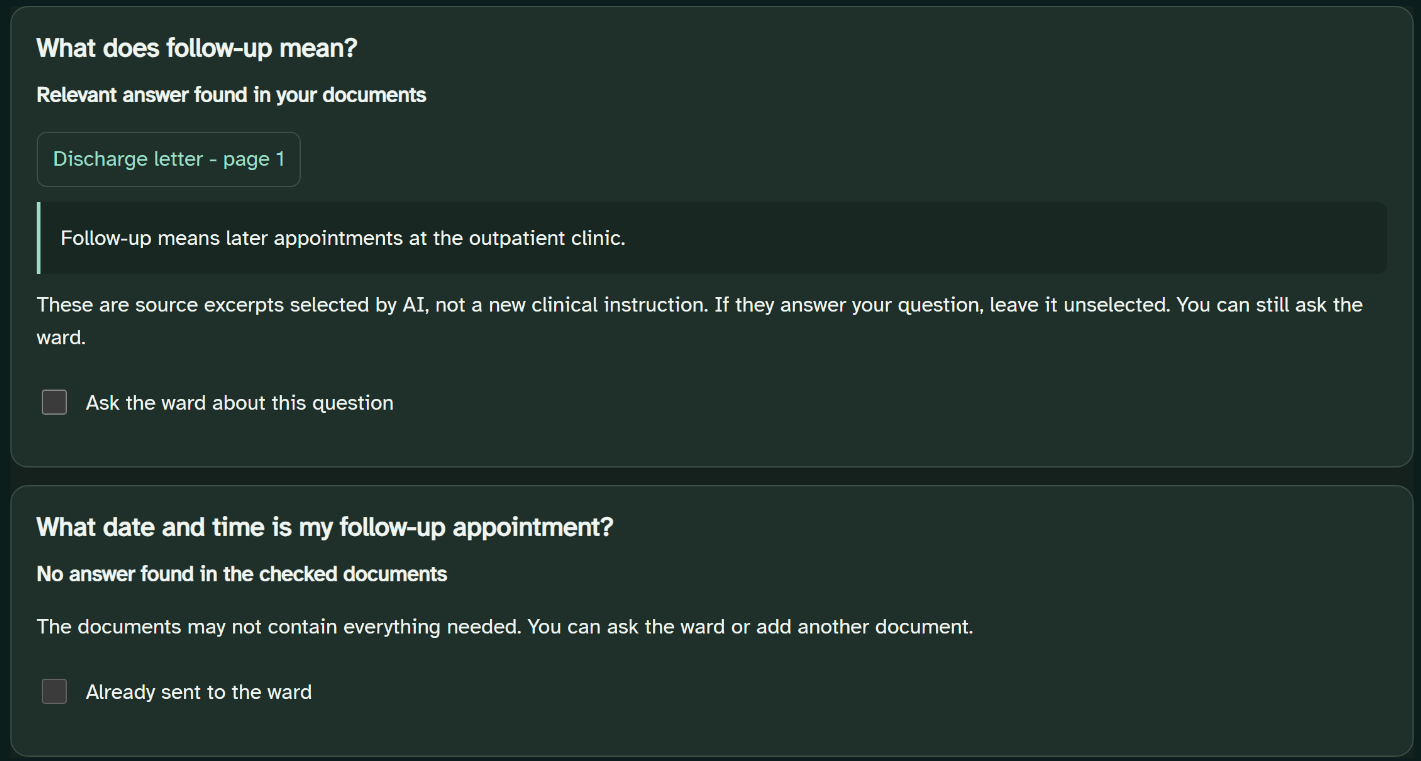
Selected questions go to staff. Follow-ups stay together. The patient marks resolution.

The key distinction: a document held by the ward is not yet a document received by the patient.

AI BEFORE CONTACT

Ask the documents before asking the ward

The patient chooses what still needs a human reply.



Already present

Read the cited passage. Send only if you still need clarification.

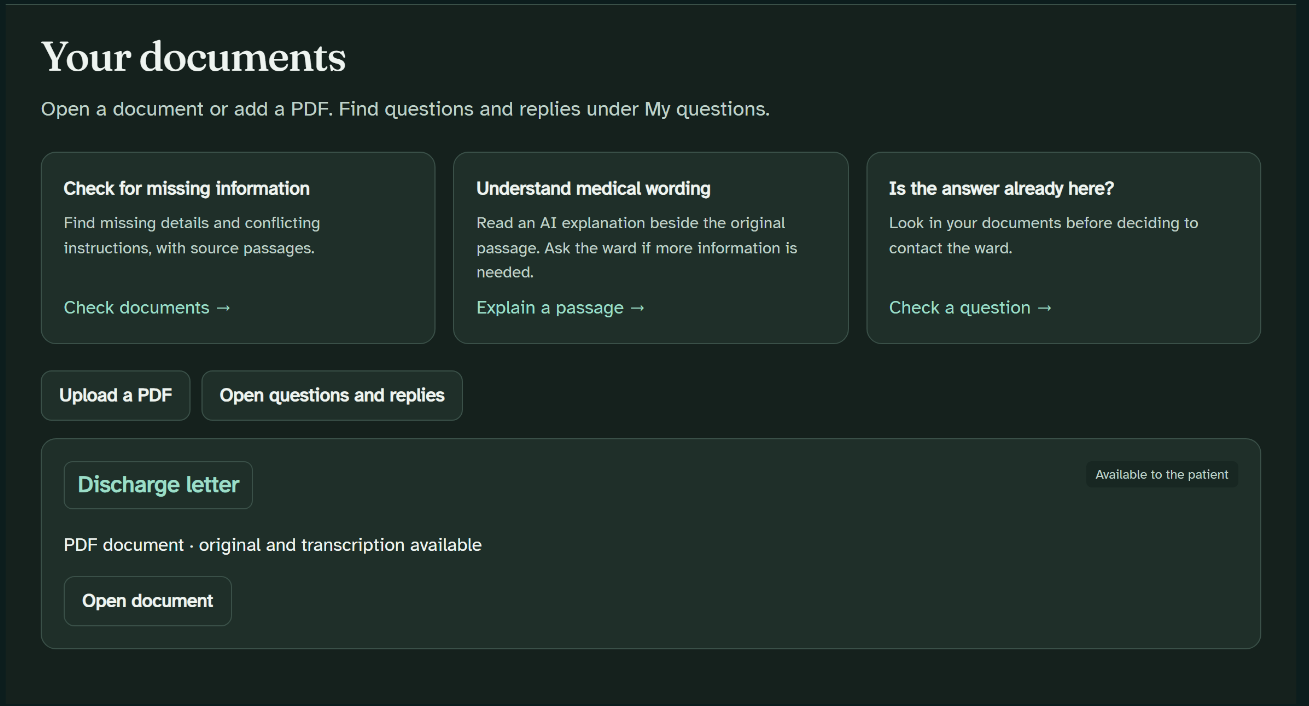
Still missing

Forward only the selected question, in context.

THE WORKING PRODUCT

AI help is part of the document workflow

Check the packet, ask about its wording, or look for an existing answer.



Ask in your own words with a quoted passage. Contacting the ward is a separate choice.

TECHNOLOGY

Google Gemma-4-31B-IT through AMD

Token Factory powers the working demo. Source checks and human actions surround the model.

Task	Reasoning	Publication / control
Scanned PDF transcription	None	User checks every page
Missing or conflicting instructions	Medium	Facts, findings, conditional conflict check
Answer in patient documents	Medium	Exact excerpts; patient chooses what to send
Ward reply draft	Medium	Staff edits and sends
Prepared clinical explanations	Medium	Shared cache; direct patient access
Free-form document question	None	Question + exact quote + reviewed document

Python + SQLite + PDFium + Railway. Event FAQ: Token Factory API qualifies; GPU credits are optional.

MEASURED EVIDENCE

High recall, with patient false alerts to reduce

Frozen English synthetic documents. Google Gemma medium. Errors retained.

Task / pipeline	Cases	Exact cases	Precision	Recall
Patient document gaps / v7	100	73.00%	64.15%	94.44%
Ward request handling / v2	100	97.00%	98.31%	96.67%

Exact = the whole expected finding set matches.
Precision = correct findings among alerts. Recall = expected findings recovered.

Includes 4 patient acquisition controls and 20 deterministic ward gates. Related variants are clustered.

Not clinical validation or an evaluation of every integrated UI feature.

BUSINESS VALUE

Count the staff time released by each case

Illustrative annual ward scenario. The assumptions must be tested in a pilot.

Assumption / calculation	Illustrative amount
10,000 cases x 2 net min x EUR30/hour	EUR10,000 capacity value
AI EUR0.02 + other EUR0.10 per case	EUR1,200 variable budget
Fixed operating / integration budget	EUR5,000 per year
Capacity value minus total budget	EUR3,800 per year

1.24 min

net minutes per case
to break even.
2 minutes would release
333 staff hours a year.

Buyer hypothesis: hospital discharge teams. Pilot metric: net minutes per case, including review and false alerts.

Two contributions, one economical workflow

Gemma supplies the model capability. AMD supplies the live inference service.

Model and workflow

Bounded source text, task-specific prompts, selective calls and validated outputs keep consumption visible and controllable.

Provider execution

The demo runs real image and text requests through AMD. Shared capacity and free quota are monitored; failures stay visible.

10k input + 2k output tokens: about $0.00168 at the CoreWeave / OpenRouter reference tariff.

Observed 17 Oct 2026. OCR and retries add consumption. AMD quota coefficients do not establish a paid-price saving.

HUMAN CONTROL

Safety and the boundary for a hospital pilot

AI helps people inspect and understand documents. Clinical decisions remain with professionals.

Uncertainty	Invalid output is withheld. Missing clinical context goes back to people.
Source review	Exact quotes and revisions are checked. Meaning still needs human judgement.
Human choice	Patients choose contacts and resolution. Staff review drafts and share explicitly.
Health data	Only packaged examples online. Verified identity and healthcare data terms are pilot prerequisites.

Pilot: independent clinical review, patient understanding and net staff minutes including false alerts.

The handoff is the product

Document possession, document delivery and patient understanding are different states.

Before a question is sent

Check the packet before discharge.
Explain unfamiliar medical wording.
Find answers in the documents
the patient actually received.

When a person is needed

Send only the remaining questions.
Keep replies and follow-ups together.
Let staff answer directly.
Let the patient confirm resolution.

TRY IT

Follow one question from source to resolution

Open the guided cases and inspect the public code.

afterward-demo.up.railway.app

Case 1 Find an answer, then follow the remaining question.
Case 2 Check a packet before sharing it.
Case 3 Read a scan, understand it and check its instructions.

https://github.com/AlessandroFlati/AfterWard